Parv

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parv68 Parv Ruhil

Professional Summary

Systems-oriented Full Stack Developer specialising in **embedded runtimes, open-source infrastructure, and production web applications**. Author of **4 shipped open-source projects** in Rust, Go, and TypeScript — spanning authorization engines, encrypted sync runtimes, developer tooling, and cryptography. Proven track record delivering **multi-tenant SaaS products** with real-world deployments, **40%** backend performance gains, and **35%** user engagement improvements. Comfortable owning projects end-to-end: architecture, implementation, deployment, and iteration.

Projects & Experience

Open Source Developer — Independent *Apr 2026 – Present*
Rust, Go, TypeScript, WebAssembly, PostgreSQL, SQLite, gRPC, Docker

- Designing and shipping **production-grade open-source infrastructure** focused on reliability, portability, and developer experience, with **16+ releases** shipped across projects.

DevBoxOS — Universal Development Sandbox *Go, gRPC, Docker, Protocol Buffers*

- Built a local-first dev environment manager replacing Docker Compose + shell scripts with a single `devbox.yml` and one command — **16 releases shipped** with cross-platform support (Windows, macOS, Linux).
- Designed a **gRPC-based CLI/daemon architecture**: cobra CLI ↔ engine daemon ↔ Docker SDK, with graceful daemon fallback to direct Docker API calls when engine is unavailable.
- Implemented auto-detection for **12+ languages and services** (Node, Go, Rust, Python, Java, Ruby, PHP, PostgreSQL, MySQL, Redis, MongoDB, Docker) by scanning source files and config patterns.
- Built **encrypted secrets management** via `age` (X25519 + ChaCha20-Poly1305), environment snapshots (save/export/import/restore), hot reload, real-time resource monitoring, and Docker Compose import/export.

Aegis — Embedded ReBAC Authorization Runtime *Rust, TypeScript, Go, Python*

- Built an embedded, relationship-based authorization engine (ReBAC) inspired by Google Zanzibar — runs **in-process with zero separate infrastructure**, eliminating network-latency permission checks entirely.
- Designed for sub-millisecond permission evaluation through in-memory graph traversal and parallel branch execution.
- Built language bindings via **NAPI** (Node.js), **PyO3** (Python), and **CGo** (Go), enabling the Rust core to be used natively from 4 language ecosystems with zero performance overhead.
- Supports **SQLite** (WAL mode), **PostgreSQL**, **MySQL**, and **RocksDB** storage backends; includes multi-tenancy, GDPR compliance APIs, full audit trail, and explain/trace API.

VaultSync — Local-First Encrypted Sync Runtime *Rust, WebAssembly, TypeScript, React*

- Built a **CRDT-native** (Yrs) sync engine making applications work fully offline and sync automatically — every UI write is **sub-millisecond** because it hits local storage, never a remote server.
- Implemented **zero-trust E2EE**: X25519 key pairs per replica + ChaCha20-Poly1305 symmetric encryption; coordinator stores only encrypted blobs, enabling zero-trust deployment architectures.
- Designed automatic crash recovery and synchronization resilience across browser tabs using heartbeat monitoring and leader election.
- Shipped React SDK (`useQuery`, `useSyncStatus`, `useVaultSyncMutations`) and integrated **OpenTelemetry** tracing + Prometheus metrics across every operation.

Full Stack Developer Intern — Ruhil Future Technologies *Feb 2025 – Dec 2025*

- Collaborated with a cross-functional team of **2 members** to design and enhance ERP features, reducing bug reports by **32%**.
- Optimised backend APIs and database queries, improving response times by **40%** for production systems.
- Integrated AI-powered features including automated quiz generation and student performance analytics, boosting platform adoption by **28%**.

- Conducted client requirement analysis and delivered bi-weekly demos, achieving **100% client satisfaction rate**.

Gyansetu School ERP *Full Stack, Multi-Tenant Architecture, Role-Based Access*

- Sole developer for **80%+ of the codebase**, delivering a production-ready ERP currently serving **real schools with active users**.
- Designed and developed a **full-scale multi-tenant School ERP** supporting multiple schools in isolation, ensuring complete data separation and security.
- Implemented **role-based logins** for Administrators, Teachers, Parents, and Students with tailored dashboards and permissions.
- Built comprehensive modules: student profile management, teacher diary, timetable, attendance, transport, salary, fees management, expense tracker, and parent portal.
- Automated fee generation and syllabus tracking per school, increasing admin efficiency by **25%**.
- Developed AI-powered recommendation systems and adaptive quizzes, improving student engagement by **35%**.

Technical Skills

Systems & Infrastructure: Rust, Go, WebAssembly (WASM), gRPC, Protocol Buffers, Embedded Runtimes, CRDTs
Security & Cryptography: AES-256-GCM, X25519, ChaCha20-Poly1305, PBKDF2, E2EE, LSB Steganography
Languages: JavaScript, TypeScript, SQL, HTML5, CSS3
Frameworks & Libraries: React.js, Next.js, Node.js, Express.js, TailwindCSS, Tauri
Databases: PostgreSQL, MongoDB, MySQL, SQLite
ORM / Query: PrismaORM, Drizzle
DevOps & Tools: Docker, Git, GitHub, GitHub Actions, Postman, Railway, Vercel, AWS, Cloudflare
Observability: OpenTelemetry, Prometheus Metrics, Structured Logging
Concepts: ReBAC / RBAC, Multi-Tenant Architecture, Local-First Architecture, REST APIs

Education

Bachelor of Computer Applications (BCA) — Maharshi Dayanand University, Rohtak *Aug 2023 – Jun 2026*
Relevant Coursework: Data Structures, Database Systems, Computer Networks, AI & ML Basics

Achievements & Highlights

- Authored and shipped **4 production-grade open-source projects** in Rust, Go, and TypeScript — spanning authorization engines, E2EE sync runtimes, developer tooling, and cryptography, with **16+ releases**.
- Built a **multi-tenant School ERP** (sole developer for 80%+ of codebase) currently serving active schools, delivering **40% backend latency reduction** and **25% admin efficiency improvement**.
- Built and shipped DevBoxOS, VaultSync, Aegis, and StegaShare across Rust, Go, and TypeScript ecosystems.